Absolute values

1 Consider the number line below:

a How far is the number 3 from 0?

b Hence, state the value of $|3|$.

c How far is the number -7 from 0?

d Hence, state the value of $|-7|$.

2 Describe what the absolute value of -5 represents on a number line.

3 State whether the following have a value of $|-37|$:

A Both 37 and -37 **B** -37 only. **C** 37 only.

4 Determine whether the following have a value of 37:

A Both 37 and -37 **B** -37 only. **C** 37 only.

5 Evaluate the following:

a $|6|$
b $|65|$
c $|3341|$
d $|-3|$
e $|-67|$
f $|-155|$
g $|11|$
h $|-21|$
i $|-1000|$
j $|-19|$
k $|-756|$
l $|8|$

6 Evaluate the following:

a $|-2|+|5|$
b $|-2+5|$
c $|8-(-6)|$
d $|8|-|-6|$
e $|-3|+|-7|$
f $|-3+(-7)|$
g $|8-9|+|-6+4|$
h $|9-11|-|-8+5|$

7 Evaluate the following:

a $|-3| \times |4|$

b $|-3 \times 4|$

c $|10 \times (-2)|$

d $|12| \div |-4|$

e $|-8| \times |-4|$

f $|-8 \times (-4)|$

g $|6-8| \times |-7+4|$

h $\left|\dfrac{5-7}{-4+(6)}\right|$

i $\left|\dfrac{18-10}{3-7}\right|$

j $\dfrac{|6-12|}{3-4}$

8 If $|a| = 6$, what values could a be equal to? Explain your answer.

9 Find the possible values of x for each of the following:

a $|x| = 9$

b $|x| = 4$

c $|x| = 2$

d $|x| = 7$

e $|x| = 11$

f $|x| - 8 = 0$

g $|x| - 10 = 0$

h $|x| - 3 = 1$